

Re-analysis Summary Report for Peer-evaluation

Paper ID: Hurst_EvoHumanBehavior_2017_yypJ

Re-analyst's ID: 8gs27

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
... those with a faster life strategy report greater levels ... of psychopathology (p. 1.)

The corresponding article

Here you can find the original data and article: https://osf.io/vhmgc/?view_only=5c6bedb36b2549a88ac137d6c746bcb8

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

I first recalculated the total (sum) scores of the variables of interest (i.e, Mini-K, HKSS, and DSM5), then I checked whether the descriptives I got for these variables matched those values reported in the paper. Then I conducted a partial correlation between the sum scores of Mini-K ($M = 14.62$, $SD = 11.25$), HKSS ($M = 87.35$, $SD = 13.64$), and DSM5 ($M = 40.10$, $SD = 14.57$) while controlling for age (as this variable was considered of importance by the original authors). This analysis was chosen to test the hypothesis that a slow life strategy (i.e., having fewer children but investing more in the upbringing) compared to a fast life strategy (i.e., having more children but investing less in the upbringing) is associated with lower levels of psychopathology. To test this hypothesis, I would have liked to compare two groups (those with a fast life strategy and those with a slow life strategy) rather than computing a correlation, as was done by the original authors. However, some research on the questionnaires, that I was not familiar with, suggest that a correlation is commonly used to make claims about fast vs. slow life strategies. That's why I used correlation as well. The results show that a slower life strategy, as measured by the Mini-K as well as the HKSS, is indeed associated with lower levels of psychopathology ($r[136] = -.51$, $p < .001$, and $r[136] = -.41$, $p < .001$, respectively). Please note that for the Mini_K as well as the HKSS higher scores are associated with a slower life strategy.

The analyst's conclusion: A slower life strategy is associated with lower levels of psychopathology.

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

You should report the Mini-K measure instead of HKSS for life history strategy.

Here, the analyst reported the following steps and results of the analysis:

The hypothesis being tested is that a slow life strategy (i.e., having fewer children but investing more in the upbringing) compared to a fast life strategy (i.e., having more children but investing less in the upbringing) is associated with lower levels of psychopathology. To test this hypothesis, I compared two groups (those with a fast life strategy and those with a slow life strategy) and conducted a t-test. I first created two groups based on the sum score of the Mini-K measure. Negative values are associated with a fast life strategy and positive values with a slow life strategy. Therefore, every participant with a sum score equal to or smaller than -1 was assigned to the fast life strategy group (value = 0), and every participant with a sum score equal to or greater than 1 was assigned to the slow life strategy group (value = 1). There were two participants with a sum score between -1 and 1. Those were excluded from analysis because their score did not indicate a clear preference for a slow or fast life strategy. For total DSM5 scores, the dependent variable, Levene's test accepted the null hypothesis of equal population variances, $F(1,134) = 0.165$, $p = .686$. However, a Shapiro-Wilk test showed that the distribution of the data in the slow life strategy group were not normally distributed ($W = 0.846$, $p < .001$). Therefore, I performed the non-parametric Mann-Whitney U-test. According to this test, there is a significant difference in level of psychopathology between participants with a fast ($Mdn = 60.50$) and a slow life strategy ($Mdn = 33.00$; $W = 1671.00$, $p < .001$).

Analysis language/software/code (optional to check):

Task 1: JASP

Task 2: JASP

Analysis codes: <https://osf.io/tqauz/>